

What is driving the signal generators market?

Electronics designing and assembly necessitates the use of various types of electronic test equipment ranging from uncomplicated and economical to perplexing and fancy-like ATE, for instance. On the premise of different applications, signal generators are generally used in the development, designing, testing, troubleshooting and repairing stages of electronic devices by different end-users such as aerospace and defence, automotive, electronics manufacturing, telecommunications as well as the education sector with the communications industry continuously driving the growth of this market.

oscillator (LO) is now more stabilised, and this provides high performance and accurate measurement for many mission-critical measurements.

Talking about the architectures used in signal generators and how they help a design engineer, Manish Joshi, deputy CEO, Scientech Technologies Pvt Ltd informs, "In signal generators, for lower frequencies, DDS-based architecture is used as it is lower in cost and best in performance as compared to conventional discrete-based approach. For a higher frequency application like wireless communication, more than one channel is used with software connectivity for downloading different waveforms based on wireless communication standards."

He further adds, "It will help a design engineer to get smooth and high-frequency resolution signal and to design him/her own customised signal.

Programmable versus basic signal generators

"Signal/function generator with software allows a user to easily create his own waveform and edit and download complex waveforms using the wave editor, whereas with the basic signal generators the user can access only factory-defined waveforms," informs Joshi.

The basic signal generators are still being used for primitive tests that are being conducted on some of the electronic components that are designed, but when it comes to the high-speed verification/validation or the high-speed manufacturing and testing, the days of the stand-alone box-type signal generators are gone, believes Mohanram.

T. Anand, principal consultant, Knewron, says, "The buzzword now



3MHz handheld synthesised function generator

is programmable; hardcoded or fixed physical signal generation is thing of past. More so, it is need of today's technologically evolved world." On another note, Tripathi believes, "Industry looks for physical signal generators as they are popular. But both categories have own merits and final decision is made based on performance, application, cost and ease of operation."

New initiatives to help design and test better

FPGA-based instruments. From signal generators' standpoint, there are a lot of people developing and working on specific IPs that can be implemented on the FPGA and the signal generator, shares Mohanram. He says, "There is a good level of standardisation of these IPs, and hopefully we will be able to leverage all of these across platforms of signal generators. There are different vendors having different FPGAs on their signal generators; some of them are 'open' and some of them are not."

Different vendors are coming up with open signal generators, and these

IPs can be quickly leveraged by these open FPGA signal generators. There is some work happening in terms of standardising this architecture and coming up with IPs for the same.

On a similar note, Vishal Gupta informs, "FPGA-based signal generators have been there for quite a while. Some of the Agilent ARBs have in-built FPGAs where data generation comprises a sample memory which contains the sample data, a sequence memory which contains the required information for sequencing such as the sequence structure or loop counter values and a channel FPGA which combines both into a sequence-controlled sample stream."

Software tools to control the instrument. Many signal and function generators now come with a facility to be controlled from custom programs such as those built in Visual Basic (VB) or otherwise, informs T. Anand. He says, "These facilities make these generators suitable candidates for getting included in automated test equipment (ATEs) such that feedback-loop-based tests could be easily designed."

Talking about tools, Mohanram says, "There are tools coming into the marketplace which can integrate the design environment with these signal generators directly, because when you are talking about building this particular model which has to run on the signal generator, it generally happens in some kind of a design environment. So there is a good level of work happening to integrate these design environments along with software which actually hit the FPGA program on the signal generator."

Sharing his thoughts on some other new initiatives, "Waveform generator companies are providing software/tools which will help designers to design and test their devices under test (DUT) in a much more simplified way. Portable handheld function generator is available for field applications," explains Joshi.

Growth in usage

The evolving electronics industry is